

Trainings ▾

Preparatory Steps

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

This procedure describes several aftercooler arrangements.

- Outboard Aftercooler - Standard Cooling System
- Outboard Aftercooler - Low Temperature Aftercooler (LTA) Cooling System
- Center Mount Aftercooler.
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Remove air coolant plumbing. Refer to Procedure 010-002 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html)

Remove

Standard

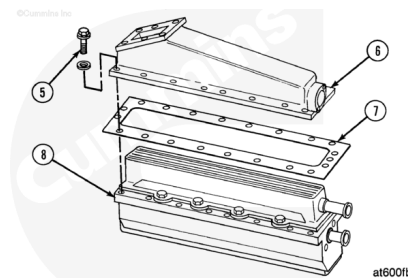
Note : This procedure is performed while the intake manifold (aftercooler housing) remains on the engine.

Remove the 21 capscrews and flat washers (5).

Note : Do not damage the gasket surfaces if it is necessary to use a wedge to separate the parts.

Remove the aftercooler cover (6). It will be necessary to move the cover so it will **not** hit the nipple on the core.

Remove the gasket (7) from the intake manifold (8).



⚠ WARNING ⚠

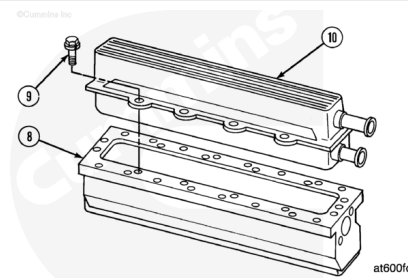
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Use a solvent that will not harm copper.



Remove the eight captive capscrews (9). Notice that the washers on the capscrews are spring-type washers.

Remove the core (10) from the intake manifold (8).

Clean the element.

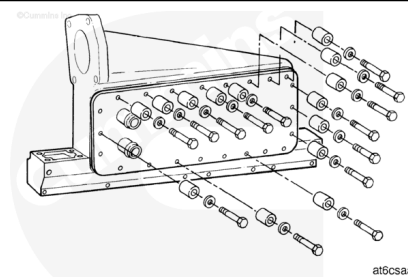
Use solvent or steam. Clean the remaining parts.

Check for damage.

LTA

Remove the capscrews and spring washers.

Remove the aftercooler element and rectangular strip seal.



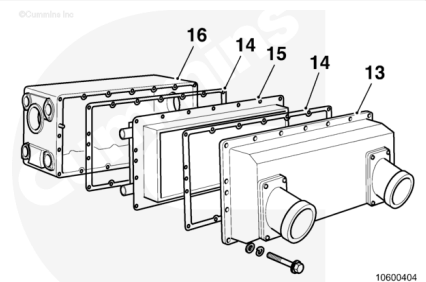
Center-Mount

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

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Remove both of the aftercooler covers (13) and gaskets (14).

Remove the aftercooler cores (15) and the gaskets (14) from the housing (16).

Clean all o-ring bores and grooves. Clean all gasket surfaces.

Use solvent and steam. Clean the parts.

Check the parts for cracks or damage.

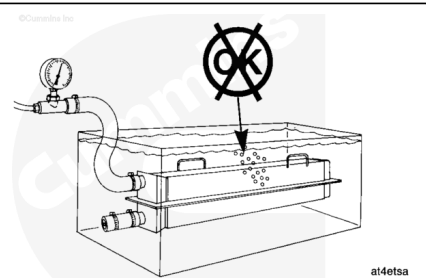
Note : If the K50 aftercooler housing does not contain cross bolts, or contains a rib in the center of the housing, it is recommended that the housing be modified for the cross bolts before it is installed again.

Refer to Procedure 010-002 (Aftercooler Assembly) in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html>)

Inspect for Reuse

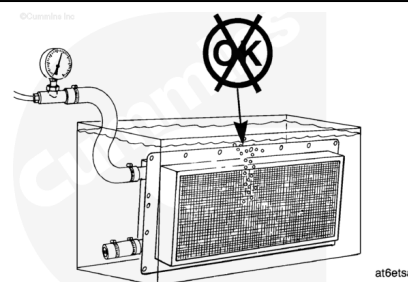
Standard

Clean and check element for reuse. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html>)

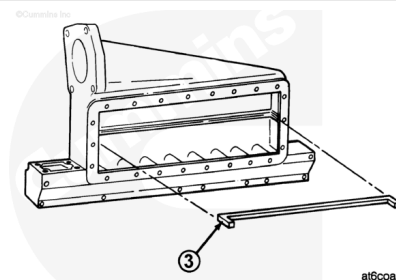


LTA

Clean and check the element for reuse.

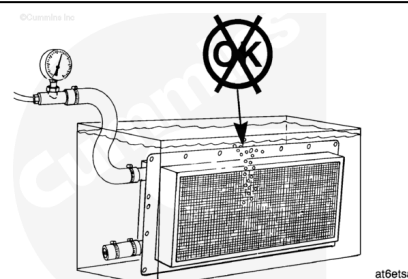


Inspect the rectangular strip seal (3). Discard the seal if it is ripped, cracked, or worn.



Center-Mount

Clean and check the element for reuse. Refer to Procedure 010-002 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html)

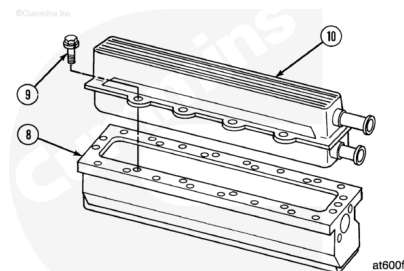


Install

Standard

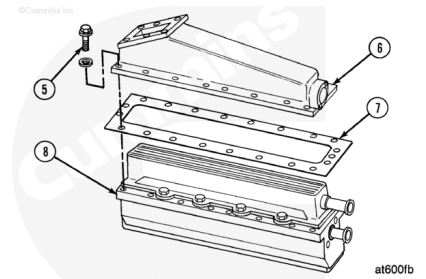
Install the core (10) in the manifold (8). Install the eight capscrews (9) that contain the captive spring washers.

Torque Value: 25 n•m [20 ft-lb]



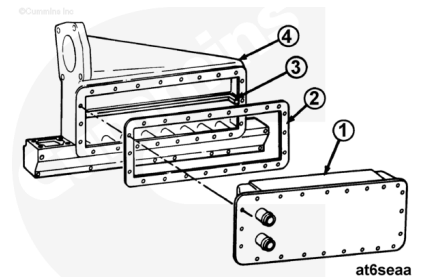
Install a new gasket (7) and cover (6). Install the 21 captive capscrews and flat washers (5).

Torque Value: 25 n•m [20 ft-lb]



LTA

Install the seal (2) into the groove in the air intake manifold housing (4). If a new seal (3) has been installed, check the length that is protruding outside the housing. This protrusion **must** be flush to 3 mm [1/8 inch] in length outside the housing at both sides. If required, clip the seal to this length. Apply a bead of sealant. Install the aftercooler core (1).

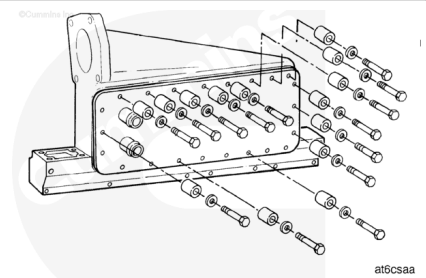


Note : All aftercooler cores are identical and are inserted into the housing with the water inlet and outlet just below the air intake connection.

Install the 3 inch capscrews and spring washers into the holes shown.

Capscrews at position 1 can **only** be placed in the rear intake manifold assemblies at this time.

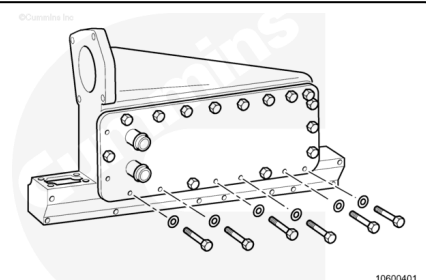
Tighten the inserted capscrews just enough to hold the aftercooler element in place.



Install the 6.75 inch capscrews and spring washers into the holes shown.

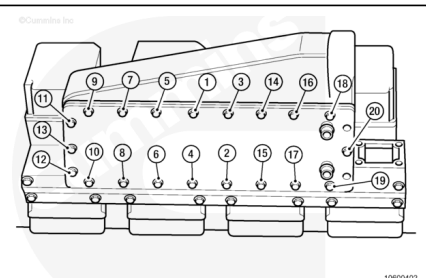
Capscrews will be inserted in the open positions either during the installation of the aftercooler plumbing or the intake manifold assembly.

Tighten all of the inserted capscrews just enough to hold the aftercooler core in place.



Use the sequence shown. Tighten the capscrews.

Torque Value: 45 n•m [35 ft-lb]



Center-Mount

Install the housing-to-core flange gasket (14). Be sure the proper holes are aligned with the roll pin.

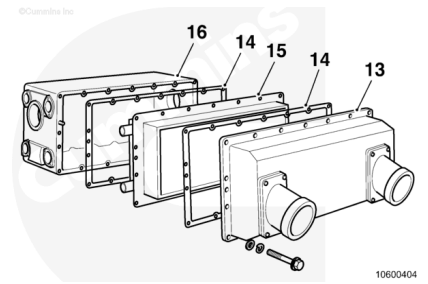
Install the cooler core (15). Install a new cover-to-core flange gasket (14) and the cover (13).

Note : Each cover contains 18 capscrews.

Install the flat washers, lock washers, and capscrews.

Torque Value: 25 n·m [20 ft-lb]

Repeat this step for the other side of the assembly.



Finishing Steps

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- Install air and coolant plumbing. Refer to Procedure 010-002 in Section 10 (</qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html>).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

